

Anirudh Rao

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EDUCATION

- 2021–2026 **Indian Institute of Technology Madras (IIT Madras)**
BTech (Honours) in Biological Engineering, BTech (Minor) in Computational Biology, and Interdisciplinary MTech in Data Science
CGPA of 9.78
- 2026–PRESENT **University of Oxford**
DPhil in Engineering Biology

ACADEMIC ACCOMPLISHMENTS

- 2026 Won the **Institute Merit Prize** for the highest CGPA across all Dual Degree (BTech-MTech) programmes at IIT Madras.
- 2026 Won the **American Express Top Achievement Award** for the best academic performance in the Interdisciplinary Dual Degree programme in Data Science at IIT Madras.
- 2025 Won the **Institute Merit Prize** for the best academic performance in the third year of the Biological Engineering programme at IIT Madras.
- 2024 Won the **Dr. Anita Mehta-Damani Prize** for the best academic performance in the second year of the Biological Engineering programme at IIT Madras.
- 2024 Selected as a **Khorana Scholar** to undertake research in biotechnology at an academic institution in the United States.
- 2023 Selected as a **Young Research Fellow** to participate in undergraduate research at IIT Madras.

RESEARCH EXPERIENCE

- 2025–2026 **Baker Lab, University of Washington**
Advisor: Dr. David Baker
Worked on improving the design success of *de novo* protein binders using contrastive learning and large-scale experimental data.
Modified an existing biomolecular structure prediction network, RoseTTAFold3, to predict binding affinity in addition to 3D structure.
- 2024 **Cong Lab, Stanford University**
Advisor: Dr. Le Cong
Worked on *in silico* optimisation of ADAR2-Cas13-mediated RNA editing in mammalian cells using protein language models.
Developed a computational pipeline to automate the analysis of killing assays to test the efficacy of NK cell-mediated cancer immunotherapy. Performed digital cytometry and survival analysis using publicly available data from TCGA.

2023–2024	Computational Systems Biology Lab, IIT Madras <i>Advisor: Dr. Karthik Raman</i> Designed and experimentally validated an <i>in vitro</i> artificial neural network for universal function approximation using DNA molecules.
2022–2025	iGEM IIT Madras <i>Team Lead (2024–2025):</i> Developed a co-culture of engineered silicon-solubilizing bacteria and diatoms for <i>in situ</i> resource utilization during Mars missions. Created Astrolabe, the first software tool for chassis recommendation in space synthetic biology (Gold Medal , iGEM 2024). <i>Lab Head (2023–2024):</i> Constructed and characterized a library of synthetic ribosome binding sites (RBS) for use in recombinant <i>Lactococcus lactis</i> . Built a machine learning model for prediction and optimization of gene expression by RBS engineering (Gold Medal , iGEM 2023). <i>Lab Member (2022–2023):</i> Designed protocols for the directed evolution of yeast cytosine deaminase to enhance stability for enzyme prodrug cancer therapy (Scientific Contribution Award , iDEC 2022). Constructed genetic circuits in <i>E. coli</i> to produce bacterial cellulose from whey water for transdermal insulin patches (Grand Prize , iGEM Indian League 2022).

TEACHING EXPERIENCE

2023–2024	Teaching Assistant for Dr. K. Subramaniam’s course on Biochemistry as part of NPTEL, the National Programme on Technology Enhanced Learning
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PUBLICATIONS

Dinesh, R. K., Wang, X., Mohammad, I. A., Gunasekaran, P., Stikloraitis, K., Villafuerte, J. R., **Rao, A.**, Hernandez-Lopez, R. A., Sunwoo, J. B., & Cong, L. (2026). Surfaceome CRISPR activation screening uncovers ligands regulating tumor sensitivity to NK cell killing. *Nature Communications*, 17(1). doi: 10.1038/s41467-026-76374-x

SKILLS

PROGRAMMING AND ML	Python, Java, C, MATLAB; ML with TensorFlow, Keras, PyTorch, Scikit-Learn; MLOps with Airflow, Spark, Docker, Git, FastAPI, DVC, Prometheus, Grafana, MLflow; HPC with Slurm
COMPUTATIONAL BIOLOGY	Molecular dynamics (VMD, GROMACS); COBRA flux balance analysis; RNA-Seq in R (Seurat); network modelling (Cytoscape, NetworkX); <i>de novo</i> protein design (RFDiffusion, ProteinMPNN)
MOLECULAR BIOLOGY	Plasmid isolation, agarose gel electrophoresis, cloning, Gibson assembly, PCR, qPCR, heat-shock transformation, SDS-PAGE